

The role of inhibitory control in shoot/don't-shoot decisions

Adam T Biggs¹  and Kyle A Pettijohn²

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Abstract

Previous work has demonstrated a link between cognitive abilities, specifically inhibitory control and lethal force decision-making performance. However, many previously used approaches to simulating shoot/don't shoot scenarios have lacked ecological validity. There is a need to investigate how inhibitory control impacts shoot/don't decisions using realistic simulations to better translate the findings to military and law enforcement settings. This study used multiple cognitive control tasks incorporating discrete judgements in go/no-go and stop signal tasks as well as subjective judgements in go/no-go tasks with both colour stimuli and emotional faces. These combined tasks provided a comprehensive evaluation of inhibitory control abilities. To ensure ecological validity in shooting performance, existing military training scenarios incorporated realistic weaponry and aiming behaviours across different shoot/don't-shoot simulations. The inhibitory control battery identified five principal components from the various tasks, including: stopping ability, response speed, emotion detection, colour detection, and emotional biases. These principal inhibitory control components were entered into hierarchical linear regressions with the dependent variables of unintended casualties inflicted and lethal rounds fired, respectively. Stopping ability better predicted the likelihood of inflicting an unintended casualty, whereas response speed better predicted the number of lethal rounds fired. These regression models included baseline metrics of marksmanship and shots fired, which supports a role for inhibitory control above and beyond basic shooting abilities or strategy. These collective findings provide mechanistic support for the relationship between inhibitory control and errors in shoot/don't-shoot decision-making while using realistic military training scenarios.

Keywords

Inhibitory control; response inhibition; lethal force; shoot/don't-shoot; military; law enforcement

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Despite the critical life-or-death implications, lethal force decisions can occur in mere seconds—or less. Specifically, shoot/don't-shoot decisions might align better with lower-order cognitive processes of perception, attention, and object recognition, whereas tactical planning and rules of engagement might align better with higher-order processes of problem-solving and reasoning. This interpretation is supported by research demonstrating how lower-order cognitive processes such as attention and perceptual judgements can be altered by wielding a weapon (Biggs et al., 2013; Witt & Brockmole, 2012; Witt et al., 2020). A critical implication is that these cognitive processes, while occurring extremely fast, can be affected by a complex and wide array of situational factors and individual differences. Moreover, these cognitive functions are not always the primary focus of lethal force research as many other variables often receive more attention.

Previous studies have often focused upon creating realism through increased anxiety to simulate stress during a

use-of-force encounter (Nieuwenhuys et al., 2012; Nieuwenhuys & Oudejans, 2010, 2011; Oudejans, 2008). Creating a realistic environment is an important component of lethal force training (Hoff, 2012), but such an environment also requires that the scenario faithfully replicates the decision-making process. This latter requirement mandates adequate representation of the perceptual judgements and cognitive processes that would be incorporated into a real-world threat assessment. Unfortunately, these judgements and processes could be biased by some artificial aspects of tasks during lethal force training. For example, shooters can become biased to make shoot/don't-shoot

¹Naval Special Warfare Command, Coronado, CA, USA

²Naval Medical Research Unit Dayton, Dayton, OH, USA

Corresponding author:

Adam T Biggs, Naval Special Warfare Command, Wright-Patterson AFB, OH 45433, USA.

Email: adam.t.biggs.mil@socom.mil

decisions due to unintentional cues present during training, such as how re-using targets can create a salient difference between hostile and non-hostile targets (e.g., bullet holes; Biggs, Pistone, et al., 2021) or how repetition creates contextual learning among lethal force stimuli (Biggs, Huffman, et al., 2021; Biggs, Pettijohn, & Blacker, in press). These subtle cognitive influences can compromise decision-making such that the choices made during training do not align with choices made in the field. This potential also underscores the need to understand how various cognitive processes might impact a lethal force decision.

Previous work has begun to investigate how different facets of cognition are involved in shoot/don't-shoot decision-making. Foremost, the decision to shoot is based on an accumulation of perceptual evidence in the milliseconds prior to the behavioural response (Pleskac et al., 2018). This period provides the opportunity for individual biases to alter the perception of someone as more or less threatening (Correll et al., 2002, 2007; James et al., 2014). However, while shooter biases have been studied more thoroughly in previous research, recent evidence has suggested a prominent role for inhibitory control in shoot/don't-shoot decision-making errors. Inhibitory control broadly describes a construct encompassing the ability to inhibit or suppress goal-irrelevant information (Tiego et al., 2018). Subcomponents address particular processes, such as response inhibition to suppress prepotent motor activity or attentional inhibition to suppress distracting information. The inability to inhibit responses in a sustained attention task has been linked to increased shooting errors using correlative evidence or task manipulations (Wilson et al., 2013, 2015). Additional evidence further supported a causal link between this cognitive ability and lethal force errors given that inhibitory control training has been successful in reducing the chance of inflicting unintended casualties (Biggs et al., 2015; Hamilton et al., 2019).

While this early work has demonstrated that inhibitory control plays some role in lethal force decision-making, there is evidence that not all inhibitory control tasks are created equal. Task-dependent distinctions raise questions about the underlying mechanism assessed as subtle task variations address different functional processes (Criaud & Boulinguez, 2013; Eagle et al., 2008; Raud et al., 2020). Go/no-go tasks may reflect automatic inhibition that depends on consistent mapping between a stimulus and response, whereas stop signal tasks may reflect controlled inhibition that requires top-down intervention (Verbruggen & Logan, 2008a). In addition, different go/no-go tasks may not elicit the same prepotent motor response due to factors, such as stimulus interval and trial ratios (Wessel, 2018; Wilson et al., 2016). These task-dependent differences affect whether the measurement captures learned stimulus-response associations versus the volitional ability to withhold prepotent motor activity—or both—within the

larger concept of inhibitory control. Other factors not specific to inhibitory control might further complicate application to use-of-force scenarios. For example, lethal force encounters are likely to involve complex emotions, which have been shown to impair inhibitory control (Pessoa et al., 2012; van Peer et al., 2019; Verbruggen & De Houwer, 2007; but see also Littman & Takács, 2017). These nuanced aspects of inhibitory control, while established in the basic science literature, have not been investigated in the context of lethal force decision-making.

Another issue concerns the concurrent perceptual and motor processes involved in using a firearm. No study has fully addressed the interplay between cognitive and sensorimotor processes involved in using a weapon. For example, a computer-based go/no-go task will likely have participants responding to single stimuli with a finger on the response key, which does parallel a simple threat/non-threat decision made for a single target. This approach is suitable when the goal is isolating the cognitive processes involved in threat assessment to maximise experimental control. However, realistic shooting scenarios incorporate more complex and diverse components as the participant must locate a possible hostile target, make a threat assessment, use appropriate sensorimotor control to aim the weapon, and finally initiate a smooth trigger squeeze to ensure the weapon fires accurately. Each additional complication of sensorimotor control to information priming could contribute to shoot/don't-shoot errors (Taylor, 2020). As such, there are questions about how inhibitory control relates to shooting performance in a more complex or scenario-based simulation. On one hand, additional cognitive steps and sensorimotor complications could create a burden that reduces attentional capacity and leads to more errors, creating a potentially larger role for inhibitory control. On the other hand, previous evidence has demonstrated that increased response complexity reduces errors on a response inhibition task (Head et al., 2020). The more complex motor response of aiming and firing a weapon rather than pressing a key might minimise the relationship between inhibitory control and shoot/don't-shoot errors. In either case, we would assert that there is a need to study the role of inhibitory control in lethal force decision errors using ecologically valid shooting scenarios.

The current investigation explored the relationship between shoot/don't-shoot performance and inhibitory control. Errant behavioural responses could be measured through unintended casualties in the shooting simulation, whereas appropriate behavioural responses could be measured through lethal shots fired upon hostile targets. A high-fidelity, military-grade shooting simulator was used to collect measures of shooting performance during training scenarios that incorporated realistic lethal force decision-making. These scenarios provided high ecological validity and, more importantly, integrate the fundamental perceptual and motor processes of weapon handling into

the outcome. Shooting variables were also collected and used as control variables to assess whether inhibitory control contributes to performance independent of the functional ability to use a firearm. A novel contribution of this work thus examines the role of inhibitory control on shooting performance above and beyond shooting performance variables such as marksmanship or shots fired. As such, the methodology represents an improvement over previous studies using computer-based tasks or gaming simulations.

To measure inhibitory control, a battery of cognitive tasks assessed multiple aspects of response inhibition. These assessments included go/no-go tasks to incorporate elements of automatic inhibition as well as stop signal tasks to measure the ability to stop-initiated responses (Donkers & Van Boxtel, 2004; Eagle et al., 2008; Nieuwenhuis et al., 2003; Verbruggen & Logan, 2008b). We also developed a novel, hybrid task to integrate the classic go/no-go task design with a dynamic change element, notably the potential to shift from a stop signal to a go signal. This go change is absent from many stop signal tasks, although it is highly relevant in a shooting scenario where a hostile target may attempt deception by presenting initially as a non-hostile. To measure individual predisposition to respond or withhold a response (i.e., bias in signal detection terms), colour-based and emotion-based perceptual tasks were included where some trials were clear cut go or no-go trials, but others were purposefully ambiguous. Taken together, this battery of cognitive tasks provided a wider range of assessment of inhibitory control abilities than prior research involving shooting performance.

This design allows for multiple hypotheses to be addressed in this study. Foremost, if the increased response complexity in firing a weapon versus pressing a button reduces inhibitory control errors, there should be a minimal relationship between shooting errors and general inhibitory control. If this relationship remains important even when accounting for shooting variables, there are several possibilities to explain why. Poor conflict resolution or response speed in distinguishing go/no-go stimuli could produce a higher rate of unintended casualties. This outcome would suggest a larger predictive role for tasks with static stimuli such as typical go/no-go tasks rather than stop signal tasks that incorporate a change dynamic. Another explanation would suggest that the ability to inhibit initiated motor responses best accounts for the relationship between general response inhibition abilities and unintended casualties during shooting scenarios. This outcome would be demonstrated by a larger role for inhibitory control tasks that incorporate a change dynamic to better address individual stopping ability rather than conflict monitoring during straightforward go/no-go tasks. A third explanation would suggest that individual bias in response best describes the relationship between inhibitory control and shooting errors. People would be predisposed to make shooting errors as they are biased to make fast and

frequent responses, especially to tasks such as subjectively detecting emotion in facial stimuli.

Method

Ethical statement

Participants completed this study voluntarily and signed informed consent documents prior to beginning research activities. The Naval Medical Research Unit Dayton Institutional Review Board oversaw the research under protocol NAMRUD.2017.0011. All data collection was conducted in compliance with federal regulations governing the protection of human subjects. Additional data have been previously published from this data set pertaining to a unique set of hypotheses, specifically the relationship between shooting performance on a low fidelity gaming simulator and a high-fidelity military-grade shooting simulator (Blacker et al., 2020).

Participants

There were 72 adults (age: $M=30.25$, $SD=7.67$; 53 males) who participated in this experiment on a voluntary basis. Individual participants reported normal or corrected-to-normal vision at the time of the experiment. Experimental tasks were divided between two experimental sessions with counterbalancing of tasks within and between sessions. Due to numerous factors including attrition and computer error, some participants did not complete all study tasks. If a participant completed only the computer tasks, but not the shooting tasks, their data were used in the factor analyses for the inhibitory control tasks without including their data in any of the shooting analyses. For the shooting simulations, the final data set included 62 participants who completed all study tasks (age: $M=30.02$, $SD=6.54$; 44 males).

Apparatus

Computer-based assessments were conducted using a laptop with a 15" screen. Participants were seated approximately 60 cm from the screen without head movement restrictions. Experimental stimuli and timing were controlled using the MATLAB programme (The MathWorks, Natick, MA) and the Psychophysics Toolbox extension (Brainard, 1997; Pelli, 1997).

Shooting simulations were conducted using the Indoor Simulated Marksmanship Trainer (ISMT; Meggitt Training Systems, Suwanee, GA), which is the official shooting simulator of the United States Marine Corps (Figure 1). The version used for this study was the enhanced (E) model. Each system consists of an operator station connected to a hit camera and projector. For this simulation, participants stood 15 feet from a projection screen 75" in width $\times$ 57" in height. The participant shooting line was



Figure 1. Top images depict two examples of shoot/don't-shoot scenarios as conducted using the Indoor Simulated Marksmanship Trainer (ISMT). Bottom image depicts the M9 pistol used during the experimental shooting scenarios.

marked off in caution tape, and participants were allowed to maintain whatever shooting stance they chose so long as they did not cross the shooting line. Bluefire[®] weaponry was used, which involved real firearms with internal components modified to house an infrared laser and Bluetooth transmitter. Participants used the Beretta M9 pistol for these shooting simulations. Refillable magazines contained compressed air and simulated recoil with an air burst upon firing the weapon. Participants received feedback during the video-based shooting simulations. Shot placement indicated where the last round landed and the type of shot: green indicated a miss, yellow indicated a non-lethal hit, red indicated a lethal hit, and purple indicated a false alarm where the shot hit a non-hostile actor in the simulation.

Classic go/no-go task

Participants began with an instruction screen that indicated each trial would begin with a fixation cross (+) in the centre of the screen. This fixation cross would be replaced after a random interval between 0.5 s and 1.5 s with either a purple or yellow square. Each square subtended $5^\circ \times 5^\circ$ in visual angle. Participants were randomly assigned either purple or yellow as their go colour. If the go colour appeared, participants were instructed to hit the space bar as fast as possible. If the no-go colour appeared, then participants were told not to press anything. Participants were asked to respond as quickly as possible without sacrificing

accuracy. A square would remain on-screen until response or until a timeout of 1.5 s. There were 80% go trials and 20% no-go trials during the experiment. Each participant began with a practice block of 20 trials followed by 400 experimental trials. This task produced two dependent variables: go RT and no-go accuracy. Sample stimuli are available in Supplementary Materials (Figure S1).

Hybrid go/no-go task

The structure and procedure of the hybrid go/no-go task were similar to the classic no-go task except with change dynamics introduced. Colours were switched to blue and orange rather than purple and yellow to prevent learned associations from carrying over between tasks. Participants received the same instructions and procedure as with the classic go/no-go task with the information that squares could change colour. If the square changed into a go colour from a no-go colour, then the participant should respond. If the square changed into a no-go colour from a go colour, then the participant should not press anything. Static trials remained the same colour until a response was made or the 1.5 s timeout was reached, whereas change trials included a square that changed colour 150 ms into the trial. Unlike other adaptive stop signal tasks, the change here always occurred at 150 ms if the change was to occur at all. Trial proportions were divided among 70% static go trials, 10% static no-go trials, 10% change go trials (or go-to-no-go), and 10% change no-go trials (or no-go-to-go). This trial

distribution was intended to parallel the classic go/no-go task with 80% of trials that should end in a response and 20% of trials that should not end with a response. This task produced four dependent variables: static go RT, change go RT, static no-go accuracy, and change no-go accuracy. Sample stimuli are available in Supplementary Materials (Figure S2).

Stop signal reaction time (SSRT) task

Participants began each experiment with a screen informing them that a fixation cross (+) would appear in the centre of the screen. Each fixation appeared for a random interval between 1.25 and 2.25 s, followed by a square. The square would be pink or brown, and participants were randomly assigned either pink or brown as their go colour. All squares would begin as the go colour, but participants were informed that some trials could have the square turn into the no-go colour. Their instructions were to respond as quickly as possible to the go colour and to withhold their response if the square changed into the no-go colour. Participants were instructed to respond as quickly as possible without losing accuracy.

The first block of trials was considered practice, but also served to create a feedback sound. If participants slowed more than two standard deviations beyond their practice mean, the computer would sound a warning beep. Eighty practice trials were followed by 300 experimental trials. Both practice and experimental trials included 65% go trials, where the square began with the go colour and remained the same, and 35% stop signal trials, where the trial switched to the no-go colour after the stop signal delay (SSD). A one-up/one-down staircase procedure was used to determine changes in SSD. If participants responded accurately, then SSD increased to make the task more difficult. If participants responded incorrectly, then SSD decreased to make the task easier. This approach is intended to keep stop signal accuracy around 50%. Sample stimuli are available in Supplementary Materials (Figure S3). These combined methods adhere to the most recent best practices in measuring an SSRT score (Verbruggen et al., 2019), including such factors as using a salient stop signal and confining the stop signals to a minority of trials. The percentage of stop signal trials is slightly higher than the recommended 25% (Verbruggen et al., 2019), although the instructions did inform participants not to wait and to avoid going slowly to be more accurate.

SSRT score was calculated through the integration method (Verbruggen et al., 2010, 2013). This technique used go RT for the n th trial minus average SSD score to calculate SSRT. The n th trial is determined by rank or sequentially ordering go RTs and identifying the n th trial as the one corresponding to stop signal accuracy. For example, if there were 100 go trials, and the participant withheld responses on stop signal trials 40% of the time,

then the n th trial would be the 60th fastest trial. SSRT score was the only dependent variable from this task.

Colour continuum task

Participants were informed that each trial would begin with a fixation cross (+) that appeared in the centre of the screen for a random interval between 0.5 and 1.5 s. Following the fixation cross, each participant would view shades of green and purple squares throughout the trial. Colours were chosen from the third panel of the Farnsworth-Munsell 100-Hue test for colour vision. The anchors were green and purple. Equal intervals were chosen based on the tile distances in this colour vision test such that the “mostly” green and purple variants were equidistant from the anchor. The ambiguous blue colour was chosen as being the hue equidistant from both anchors. Participants would see one square at a time with one of these five colours (green, mostly green, blue, mostly purple, or purple). Go colour was randomly assigned as green or purple, and participants were instructed to respond if the square was a shade of their go colour. If the square was a shade of their no-go colour, they were told not to press anything. Trials were evenly divided across all five colours throughout the experiment. Participants completed 20 practice trials before 300 experimental trials. Trial timeout limit was set at 1 s, and the computer would automatically proceed to the next trial. This task produced six dependent variables: pure go RT, mostly go RT, ambiguous RT, pure stop accuracy, mostly stop accuracy, and ambiguous accuracy. Both RT and accuracy were collected for ambiguous trials because there was no correct or incorrect response by design. As such, response rate is a more technically accurate description for responses on ambiguous trials, although accuracy is used for consistency in terminology. Sample stimuli are available in Supplementary Materials (Figure S4).

Face continuum task

Both the colour continuum and face continuum tasks had similar structure and procedure. However, the face continuum task used emotional faces rather than coloured squares with several adaptations. Three different actors were used to prevent overfamiliarity with a single face. Participants were instructed to respond whether the face had a threatening expression. If the face had a neutral expression, then participants were told not to press anything. The instructions indicated that responses should be made as quickly as possible without losing accuracy. Emotional faces were selected from a range of normalised faces. Pure go faces were rated as 90% angry and 10% neutral. Mostly go faces were rated as 80% angry and 20% neutral. Ambiguous faces were rated as 50% angry and 50% neutral. Mostly stop faces were rated as 20% angry

and 80% neutral. Pure stop faces were rated as 10% angry and 90% neutral. Participants completed 20 practice trials and 210 experimental trials to account for differences in counterbalancing. The practice trials included a fourth face not present in the experimental trials. This task produced six dependent variables: pure go RT, mostly go RT, ambiguous RT, pure stop accuracy, mostly stop accuracy, and ambiguous accuracy. As with the colour continuum task, both RT and accuracy were collected for ambiguous trials because there was no correct or incorrect response by design. As such, response rate is a more technically accurate description for responses on ambiguous trials, although accuracy is used for consistency in terminology. Sample stimuli are available in Supplementary Materials (Figure S5).

Shooting simulation

All participants began the shooting simulation by receiving a brief safety tutorial about avoiding injury due to simulated recoil and functioning slide mechanism while the M9 fired. Participants then received a tutorial about proper functioning of the M9, including how to operate the safety, proper sight alignment when aiming, and how to load and eject the magazine. Individuals could choose any shooting stance as long as both hands remained on the weapon and both feet remained behind the firing line. Participants were instructed to keep the same shooting stance throughout the remainder of the experiment, although they could adjust and rest their arms between individual scenarios.

A preliminary course of fire simulated flat range marksmanship. This component served as a baseline marksmanship assessment and a practice session to familiarise participants with the weapon and simulator. The marksmanship course of fire included seven exercises. The first exercise was a short-range drill with three scored shots upon a United States Marine Corps (USMC) "A" target from a distance of 13 m (15 points maximum). The second through fifth exercises all used USMC "E" targets with nine rounds per exercise. Targets would appear between 3 and 10 m either static or closing towards the shooter over 4 s (90 points maximum per exercise). The sixth exercise used USMC "E" targets as pop-up targets appearing between seven and 36 m throughout the range over a period of 60 s (130 points maximum). Finally, participants completed a long-range drill with three scored shots upon a USMC "A" target from a distance of 25 m (15 points maximum). The percentage of total points possible across all marksmanship exercises was the baseline marksmanship score for each individual.

Shooting scenarios were selected from pre-programmed training simulations available within the USMC shooting simulator. Participants were instructed to fire upon hostile targets while avoiding civilians or non-hostiles. No further

feedback or instruction was given about general rules of engagement or how to distinguish between a "hostile" or "non-hostile" target. Participants were told they would not receive feedback from the experimenter, but the shot placement information was provided (lethal hits as red, non-lethal hits as yellow, misses as green, and civilian casualties as purple). There were 33 unique video scenarios presented to participants in one of two possible pseudorandom orders. Scenarios were broadly divided into: 9 "don't-shoot" scenarios that only included non-hostile actors, 24 "shoot" scenarios that included between 1 and 5 hostiles with 0 and 2 non-hostiles, and 7 "shoot" scenarios that contained fluid hostility components, where a hostile actor initially presented as non-hostile before becoming a threat or a hostile actor ceased hostility and surrendered. Scenarios were chosen to keep a high proportion that would require firing the weapon to ensure ample prepotent motor activity (Wessel, 2018). Total number of shots fired was recorded as a dependent variable, given the potential for increased false alarms with a higher number of shots (Biggs, 2017). All shooting performance data collected from the experimental sessions were recorded initially and transcribed from video replay by a member of the research team with a second research team member confirming the results in a quality check. All shooting simulations lasted between 20 and 180 s.

Statistical analyses

All analyses were conducted with IBM SPSS® Statistics 27 software. To explore missing data, Little's Missing Completely At Random test (MCAR; Little, 1988, 1995; Little & Rubin, 1987) was performed on all cognitive task variables. The test was non-significant, $\chi^2(152)=166.13$, $p=.21$, indicating that the data were indeed missing at random. As such, missing data for cognitive tests were replaced using a multiple imputation method, which has been argued as a superior method for handling missing data than mean substitution or expectation-maximisation (Kang, 2013; Rubin, 1996; Schafer, 1999). One hundred imputations were conducted to approximate missing data, which would be a conservative approach, given that only 4.75% of data were missing across all cells (Graham et al., 2007). Multiple imputation means replaced the missing cells to construct the final data set.

When analysed together to predict shooting performance, several factors exhibited a variable inflation factor (VIF) greater than 30. Based on the context of related inhibitory control tasks and a high VIF (O'Brien, 2007), this issue indicates a high degree of collinearity among several variables and warrants a factor reduction technique. The factor analysis then serves the dual purpose of addressing collinearity among the variables and identifying common elements between the tasks. To determine whether the data available

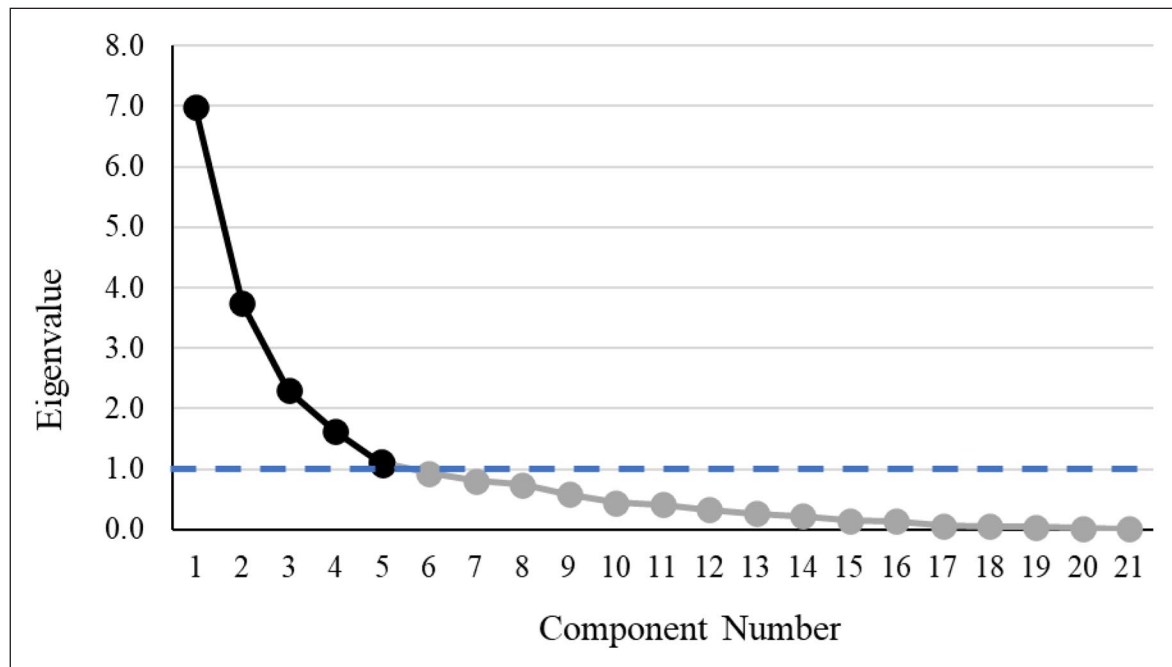


Figure 2. Scree plot of eigenvalues as measured during the principal component analysis. Dotted blue line represents an eigenvalue of 1.0, which served as the determination criteria for number of factors to be extracted.

were suitable for factor analysis, we first examined Bartlett's test of sphericity (Bartlett, 1954) and the Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy (Dziuban & Shirkey, 1974; Kaiser, 1970, 1974; Tabachnick & Fidell, 2007). For factor analysis to be appropriate, there should be a significant Bartlett's test with a minimum KMO of 0.6. The Bartlett's test of sphericity was significant, approximate $\chi^2(210)=1,405.47$, $p<.001$, with a 0.70 KMO measure of sampling adequacy. Eigenvalues greater than 1.0 were selected from the principal component analysis (Jackson, 1993; Kaiser, 1957, 1958, 1991). The varimax rotation procedure was used to create orthogonal variables that might differ as predictive factors in the subsequent shooting analyses and further ensured limited collinearity among the principal components.

Results

The results section contains several areas of analysis. Given the inhibitory control battery and novel tasks it included, the first goal was to assess which specific aspects of inhibitory control were captured across the five tasks. To do so, a principal component analysis was conducted to identify the common components between the various inhibitory control tasks and reduce extraneous or redundant predictors. With a common set of predictors, these variables were applied to the shooting simulations with the goal of predicting unintended casualties inflicted in addition to the number of lethal rounds fired against hostile targets during the scenarios.

Principal component analysis: inhibitory control tasks

The five inhibitory control tasks (classic go/no-go, hybrid go/no-go, SSRT task, colour continuum, and face continuum), had significant overlap—as expected. Twenty-one dependent variables from across the five tasks were entered into the analysis, including: classic go RT, classic no-go accuracy, hybrid static go RT, hybrid change go RT, hybrid static no-go accuracy, hybrid change no-go accuracy, SSRT Go RT, SSRT stop accuracy, SSRT score, colour pure go RT, colour mostly go RT, colour ambiguous RT, colour pure stop accuracy, colour mostly stop accuracy, colour ambiguous response rate, face pure go RT, face mostly go RT, face ambiguous RT, face pure stop accuracy, face mostly stop accuracy, and face ambiguous response rate.

Five factors were extracted based on eigenvalues greater than 1.0 (see Figure 2), which collectively explained 74.92% of the variance (see Table 1). The first factor was labelled as *stopping ability* because it incorporated classic no-go accuracy, all hybrid go/no-go variables, and the SSRT variables. Each factor represents something distinct about stopping ability orthogonal to stimulus identification and resulting response. The second factor loaded onto RT variables from the classic go/no-go task and both the colour continuum and face continuum tasks. This factor was labelled as *response speed* to identify the common element across these seven dependent variables of identifying the stimulus and responding to it. The third factor

Table 1. Principal component analysis for dependent variables collected across the five inhibitory control assessments. Values depict rotated component matrix using the varimax rotation method. All loading values below .30 have been suppressed for readability.

	Factor 1 22.09%	Factor 2 21.67%	Factor 3 13.30%	Factor 4 10.69%	Factor 5 7.19%
Classic Go/No-go					
Go RT	—	.533*	—	—	.449
No-go Accuracy	.444*	—	.376	.330	.305
Hybrid Go/No-go					
Static Go RT	.790*	.386	—	—	—
Change Go RT	.645*	—	—	—	—
Static No-go accuracy	.637*	—	.533	—	—
Change No-go accuracy	.809*	—	—	—	—
SSRT					
SSRT Go RT	.798*	—	—	—	—
SSRT Stop Accuracy	.900*	—	—	—	—
SSRT Score	-.788*	.308	—	—	—
Colour Stimuli					
Pure Go RT	—	.875*	—	—	—
Mostly Go RT	—	.815*	—	.390	—
Ambiguous RT	—	.650*	—	—	—
Pure Stop accuracy	—	—	.613	.609*	—
Mostly Stop accuracy	—	—	—	.814*	—
Ambiguous response rate	—	—	—	-.817*	—
Face stimuli					
Pure Go RT	—	.827*	—	—	.339
Mostly Go RT	—	.819*	—	—	.369
Ambiguous RT	—	.713*	—	—	-.329
Pure stop accuracy	—	—	.865*	—	—
Mostly stop accuracy	—	—	.858*	—	—
Ambiguous response rate	—	—	-.305	—	-.861*
	Stopping ability	Response speed	Emotion detection	Colour detection	Emotional bias

*indicates highest factor loading; RT: reaction time; SSRT: stop signal reaction time.

was labelled as *emotion detection* because the two dependent variables loading onto this third factor involved stopping accuracy, specifically for faces. These variables relate directly to the ability to detect the presence of emotion in faces, or lack thereof, accurately. The fourth variable was labelled as *colour detection* as a direct parallel to the emotion detection of the third factor. All three accuracy metrics of the colour continuum task loaded onto the fourth factor. Finally, the fifth factor was labelled as *emotional bias* and contained only the dependent variable of response rate for the ambiguous faces. Unlike the colour perception accuracy metrics that loaded onto a single factor, the fifth factor represents an individual bias to perceive emotion—specifically anger here—in an intentionally ambiguous face. Taken together, these five components represent the distinct underlying elements across all inhibitory control tasks included in this study.

Each individual dependent variable was recalculated into aggregate variables representing the five inhibitory control factors for subsequent analyses. To create a comprehensive variable from a diverse set of factors that

included RT, accuracy, and sometimes calculated scores (e.g., SSRT), an individual *z*-score was created for each variable. These *z*-scores were re-coded when the variable had a negative loading, such as the SSRT score onto the factor of stopping ability. The *z*-scores were then averaged together to produce a comprehensive representation of that inhibitory control component. Factors were aligned with the variable on which they most strongly loaded. The only exception was the pure stop accuracy from the colour continuum task. This result loaded equally well onto two factors, and the decision was made to align it with other variables derived from the colour continuum task with factor 4 of colour detection.

Predicting shooting performance: false alarms

A hierarchical regression was conducted on the dependent variable of unintended casualties. The first model represented shooting-related variables, including marksmanship and total shots fired. These two variables were unrelated as the marksmanship baseline assessment on a

Table 2. Hierarchical regression models for the dependent variable of false alarms.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Model summary	$R^2 = .16$ $p < .01$ *	$\Delta R^2 = .09$ $p = .01$ *	$\Delta R^2 < .01$ $p = .41$ –	$\Delta R^2 < .01$ $p = .88$ –	$\Delta R^2 = .03$ $p = .12$ –	$\Delta R^2 < .01$ $p = .65$ –
Predictors	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Marksmanship (%)	$\beta = -.257$, $p = .04^*$	$\beta = -.249$, $p = .04^*$	$\beta = -.275$, $p = .03^*$	$\beta = -.274$, $p = .03^*$	$\beta = -.227$, $p = .07$	$\beta = -.228$, $p = .07$
Total shots fired	$\beta = .349$, $p < .01^*$	$\beta = .342$, $p < .01^*$	$\beta = .363$, $p < .01^*$	$\beta = .363$, $p < .01^*$	$\beta = .341$, $p < .01^*$	$\beta = .347$, $p < .01^*$
Stopping ability	–	$\beta = -.295$, $p = .01^*$	$\beta = -.328$, $p = .01^*$	$\beta = -.322$, $p = .02^*$	$\beta = -.303$, $p = .02^*$	$\beta = -.309$, $p = .02^*$
Response speed	–	–	$\beta = .105$, $p = .41$	$\beta = .113$, $p = .42$	$\beta = .114$, $p = .41$	$\beta = .130$, $p = .36$
Emotion detection	–	–	–	$\beta = -.020$, $p = .88$	$\beta = .037$, $p = .79$	$\beta = .058$, $p = .70$
Colour detection	–	–	–	–	$\beta = -.201$, $p = .12$	$\beta = -.193$, $p = .14$
Emotional bias	–	–	–	–	–	$\beta = -.061$, $p = .65$

* $p < .05$.

Bold values indicate significant factors.

simulated flat range did not predict the number of rounds fired during the simulated lethal force scenarios, $r(60) = .15$, $p = .26$. The remaining model steps represented inhibitory control factors as entered based on the order of percentage variance explained during the principal components analysis (see Table 2). The first model explained a significant amount of the variance using the shooting-related variables, $R^2 = .16$, $F(2, 59) = 5.71$, $p < .01$, with two significant predictors in the model of marksmanship ($\beta = -.257$, $t = 2.14$, $p = .04$) and total shots fired ($\beta = .349$, $t = 2.90$, $p < .01$). Shooters were less likely to make a false alarm if they had better baseline marksmanship scores, but they were more likely to make a false alarm with each additional shot fired during the lethal force scenarios.

Only one model incorporating inhibitory control factors significantly increased the variance explained above and beyond shooting-related variables. Specifically, the model incorporating stopping ability accounted for a significant increase in the variance explained, $\Delta R^2 = .09$, $F(1, 58) = 6.71$, $p = .01$, with three significant predictors: marksmanship ($\beta = -.249$, $t = 2.16$, $p = .04$), total shots fired ($\beta = .342$, $t = 2.97$, $p < .01$), and stopping ability ($\beta = -.295$, $t = 2.59$, $p = .01$). Higher likelihood of false alarms occurred with poor marksmanship, a higher number of total shots fired, and lower ability to volitionally stop a motor response. These results underscore the importance of both marksmanship and cognitive variables in producing a shooting error. Poor marksmanship could lead to increased errors due to errant shots, and the total number of shots fired potentiates the opportunity for additional errors. Each

marksmanship aspect is distinct from the cognitive aspect of shooting errors, which involves a failure to inhibit a prepotent motor response to fire the weapon.

Predicting shooting performance: lethal hits

A hierarchical regression was conducted on the dependent variable of lethal hits fired during the scenarios. The first model represented shooting-related variables, including marksmanship and total shots fired. The remaining model steps represented inhibitory control factors as entered based on the order of percentage variance explained (see Table 3). The first model explained a significant amount of the variance using the shooting-related variables, $R^2 = .48$, $F(2, 59) = 27.24$, $p < .001$, with one significant predictor in marksmanship ($\beta = .643$, $t = 6.78$, $p < .01$). Individuals who demonstrated better marksmanship proficiency during the baseline assessment on a simulated flat range also fired a higher percentage of lethal shots during the video-based shooting simulations.

Only one model incorporating inhibitory control components significantly increased the variance explained above and beyond shooting-related variables. Specifically, the model incorporating response speed accounted for a significant increase in the variance explained, $\Delta R^2 = .05$, $F(1, 57) = 5.49$, $p = .02$, with two significant predictors: marksmanship ($\beta = .704$, $t = 7.41$, $p < .001$), and response speed ($\beta = -.236$, $t = 2.34$, $p = .02$). Lethal rounds fired were related to both marksmanship accuracy and speed on the inhibitory control tasks, where higher accuracy produced more lethality and faster speed produced more

Table 3. Hierarchical linear regression models for the dependent variable of lethal hits.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Model summary	$R^2 = .48$ $p < .001$ *	$\Delta R^2 < .01$ $p = .43$ –	$\Delta R^2 = .05$ $p = .02$ *	$\Delta R^2 < .01$ $p = .92$ –	$\Delta R^2 < .01$ $p = .90$ –	$\Delta R^2 = .01$ $p = .27$ –
Predictors	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
Marksmanship (%)	$\beta = .643$, $p < .001$ *	$\beta = .645$, $p < .001$ *	$\beta = .704$, $p < .001$ *	$\beta = .703$, $p < .001$ *	$\beta = .706$, $p < .001$ *	$\beta = .705$, $p < .001$ *
Total shots fired	$\beta = .181$, $p = .06$	$\beta = .179$, $p = .07$	$\beta = .131$, $p = .17$	$\beta = .131$, $p = .17$	$\beta = .130$, $p = .18$	$\beta = .142$, $p = .15$
Stopping ability	–	$\beta = -.075$, $p = .43$	$\beta = -.002$, $p = .99$	$\beta = -.005$, $p = .96$	$\beta = -.004$, $p = .97$	$\beta = -.017$, $p = .87$
Response speed	–	–	$\beta = -.236$, $p = .02$ *	$\beta = -.240$, $p = .03$ *	$\beta = -.240$, $p = .03$ *	$\beta = -.210$, $p = .07$
Emotion detection	–	–	–	$\beta = .011$, $p = .92$	$\beta = .015$, $p = .90$	$\beta = .056$, $p = .65$
Colour detection	–	–	–	–	$\beta = -.013$, $p = .90$	$\beta = .002$, $p = .98$
Emotional bias	–	–	–	–	–	$\beta = -.121$, $p = .27$

* $p < .05$.

lethality. This outcome underscores that lethality is related to both accuracy and speed.

General discussion

This study explored the relationship between inhibitory control and shoot/don't-shoot decisions. Specifically, multiple cognitive tasks were used to capture different elements of inhibitory control through typical go/no-go tasks, stop signal tasks, and subjective perceptual tasks designed to explore bias in colour and emotional perception. Principal component analyses yielded several common inhibitory control elements referenced in previous work, such as and stopping ability to withhold a response and response speed to identify a go signal. Stopping ability contributed to the likelihood of inflicting an unintended casualty even when accounting for shooting variables such as marksmanship ability and total shots fired. Conversely, response speed contributed to the likelihood of firing lethal shots during the shooting scenario. These differential effects contribute to the go signal and stop signal architecture used to describe the relationship between inhibitory control and shoot/don't-shoot performance. Optimal inhibitory control could improve the use of lethal force by ensuring high stopping ability to avoid inflicting unintended casualties and fast reaction times to ensure swift responses against hostile adversaries.

This evidence carries numerous advantages over prior work. Foremost, the shooting scenarios provide a high degree of ecological validity because the simulations used realistic weaponry and situations designed for military training purposes. Prior work has used various tasks

ranging from mock firearms (Wilson et al., 2013), gaming simulations (Biggs et al., 2015), or single image live fire trials (Hamilton et al., 2019). Full shooting scenarios run from initial contact to neutralising a hostile. This approach better represents contextual factors over the course of a lethal force encounter and is not isolated to recognising a single target. In addition, none of these previous studies were able to use inhibitory control assessments with such diversity and precision in conjunction with the more complex shooting scenarios. Individual tasks may be subject to strategic choices in responding as much as individual differences in response inhibition abilities (Dang et al., 2018; Verbruggen & Logan, 2008b), but assessment across multiple tasks helps identify patterns of behaviour to create a comprehensive assessment. The inhibitory control battery combined with the advanced shooting simulations provides a new level of granularity and realism in supporting a relationship between inhibitory control and shoot/don't-shoot errors. Finally, the current work highlights that the likelihood of inflicting unintended casualties is best linked to task components which quantify stopping ability. Neither response speed nor biases to detect emotion were effective predictors, whereas stopping ability maintained a significant relationship with the rate of unintended casualties while controlling for total shots fired and marksmanship. These results provide a mechanistic link to connect inhibitory control to these critical shooting errors.

From a theoretical perspective, the hybrid go/no-go task appears to capture elements of both the typical go/no-go task and the stop signal task. This task was novel compared to a stop signal task in that it incorporated both go to no-go trials (stop trials) and no-go to go trials

(dynamic go trials). These trials alter participant strategies and expectations because a stop signal is not assured to remain a stop signal. Despite the task novelty, and as evident from the principal component analysis, the hybrid go/no-go task did appear to appropriately capture elements of both the classic go/no-go task and the dynamic SSRT task. Moreover, this feature has significant implications for shooting scenarios, where a hostile initially presents as non-hostile while concealing a weapon. This possibility helps create immediate relevance to cognitive functions for military and law enforcement personnel who might encounter similar situations. Although these changes require more focused investigation about how they alter strategy and stopping latencies for inhibitory control tasks, the approach parallels other emerging evidence asking what best predicts impulsivity or inhibitory control—such as the reliability of trigger failure as a predictor versus SSRT score (Skippen et al., 2019). The potential for stop signals to become go signals also has implications for independent horse race models that will need to be addressed in future studies (cf. Logan & Cowan, 1984; Logan et al., 2015; Verbruggen & Logan, 2009) along with the validity of exploring shoot/don't-shoot decisions in either a go/no-go or two-alternative forced choice task (Gomez et al., 2007). The premise remains similar to the stop-change, albeit the individual is not initiating a separate go process from the main task.

Several limitations and future directions should be considered when evaluating these conclusions. A practical limitation involves the relationship of inhibitory control to specific shooting scenarios. A lethal force encounter carries so many different variables that is difficult, if not impossible, to address them all in a single study or simulation. Individual scenarios could present a distressed person with a loaded firearm who may pose a danger to others or themselves, and another simulation could pit the shooter against multiple hostiles with non-hostiles in the line of fire. Different scenarios such as these could emphasise different components of inhibitory control, although stopping ability appears to be an ability related to shoot/don't-shoot errors. In addition, this study did not explicitly manipulate anxiety by replicating the force-on-force elements of lethal force encounters (Taverniers & De Boeck, 2014; Taverniers et al., 2011). This anxiety-provoking element may further elevate the importance of inhibitory control by biasing the individual to shoot (Nieuwenhuys et al., 2015). The role of inhibitory control in shooting errors under higher threat conditions remains an important future direction for additional research. Future directions will also need to address the role of additional interference in a complex task such as a lethal force engagement, given that other evidence suggests a minimal role of dual-task interference on inhibitory control (Yamaguchi et al., 2012). Finally, priming and rules of engagement could be manipulated in future research.

These factors implicate individual differences rather than elements of the scenario, which can impact performance, given that priming can impact the decision to shoot (Taylor, 2020) and different individuals may not make similar responses when presented with an ambiguous threat (Biggs, Pettijohn, & Gardony, 2021). The combination of scenario-based factors, anxiety, and variable rules of engagement create a rich potential for future research to extend the findings presented here.

In conclusion, inhibitory control plays a prominent role in shoot/don't-shoot decisions. The cognitive influence remains prominent even when accounting for basic shooting performance variables such as general marksmanship capabilities or total shots fired. Stopping ability appears to be the primary factor contributing to shooting errors, whereas response speed best described effective shots fired during the shooting simulations. It is possible that emotional bias components would better depict shooting errors or a predisposition to shoot if the scenario could more readily evoke those individual biases (cf. racial prejudice; Correll et al., 2002). As a general influence, emotional biases were a non-significant predictor compared to the more dominant role of poor inhibitory control abilities. The collective evidence further emphasises the potential value of cognitive training for military and law enforcement personnel (Blacker et al., 2019) by clarifying the mechanisms underlying inhibitory control failures in shoot/don't-shoot decision-making.

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ORCID iD

Adam T Biggs  <https://orcid.org/0000-0003-3394-0691>

Supplementary material

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